

# **Automated Detection of Pork Rasher Packaging Errors and Meat Quality**

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MMA3001 — Numerical Methods and Machine Learning

Dataset 3 — Frozen-Meat Packaging Quality

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# 1. The Engineering Problem

*Explain: the engineering context; the problem being addressed; why the problem matters; the intended use of the computational solution; and the limitations of the selected scope.*

# 2. Inputs, Outputs, and Data

*Define: the input data accepted by the solution; the expected format, units and domain of the inputs; the outputs produced; the meaning and intended use of each output; and how invalid, missing or unsupported inputs are handled.*

*Address top-level inputs and outputs first, then the specific inputs and outputs of the functions you developed (docstrings and a documentation generator can assist here).*

# 3. The Computational Approach

*Explain: how the method works at an appropriate level; why it was selected; its important assumptions; its settings or parameters; how it was implemented; and its known limitations.*

# 4. The Validation Method and Results

*Explain: what was validated; why the validation method is appropriate; which metrics or acceptance criteria were used; whether the validation information influenced model selection; and what the validation cannot establish.*

*A numerical result or a visually convincing output is not sufficient evidence of accuracy.*

# 5. Comparison and Optimisation Work

*Identify and compare at least one credible alternative or baseline. The comparison may address accuracy, robustness, computational time, memory use, implementation complexity, interpretability, scalability, or suitability for the intended engineering application. The purpose is to demonstrate the final approach was selected using evidence rather than preference alone.*

*Then analyse and, where appropriate, improve performance. Make use of profilers and timing routines, and where appropriate estimates of arithmetic intensity and the number of FLOPs, to explain and justify the performance of your code. Compare the baseline solution with the improved or selected solution and justify the final design.*

| Method      | Accuracy | Runtime | Memory |
|-------------|----------|---------|--------|
| Baseline    | —        | —       | —      |
| Alternative | —        | —       | —      |
| Final       | —        | —       | —      |

Table 1: Comparison of the baseline, alternative and final approaches.

## 6. Limitations and Lessons Learned

*An unsuccessful or partially successful method can still demonstrate excellent engineering work if you validate it honestly, identify its limitations, explain what was learned, and propose well-justified improvements.*

## 7. AI-use disclosure

*Disclose all material AI use. Explain: which AI tools were used; what they were used for; the approximate level of AI contribution; why AI was used for those tasks; how AI-generated material was checked; which important decisions remained your responsibility; any errors, limitations or unhelpful suggestions produced by AI; and how AI use affected your understanding or workflow.*

*Disclosure does not replace verification — you remain responsible for all submitted code, results, claims, references and engineering decisions.*

## References

## 8. Appendix